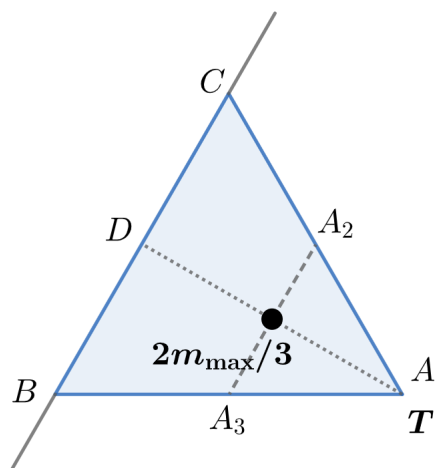


2020B F=ma Exam: Problem 6

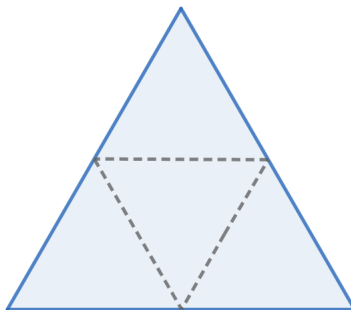
Kevin S. Huang



When a mass m is placed at the center of the triangle, $T = \frac{mg}{3}$ by symmetry, hence we have the constraint

$$T < \frac{m_{\max}g}{3}$$

Without loss of generality, consider the force T in leg A by choosing the torque axis to be BC (through the other two legs) so we only need to account for the torques from the object and T . Since the object weighs double the maximum T allowed, we see that at most it can be placed at midpoint between A and D (midpoint of BC). More generally, it has to lie outside the small triangle formed by the A vertices.



The same holds for the other two legs so the mass can only be placed in the small triangle formed by the dashed lines. Thus, the answer is A.